Open Cone (Truncated Cone / Frustum) Development Formula

Given Dimensions:

D = Bottom diameter
d = Top diameter
H = Vertical height

1. Slant Height (L)

$L = \mathrm{sqrt}(H^2 + ((D - d)/2)^2)$

2. Development Radii

Outer Radius (Ro) = $(L \times D) / (D - d)$

Inner Radius (Ri) = $(L \times d) / (D - d)$

3. Sector Angle

$\theta = 180° \times (D - d) / L$

or in radians: $\theta = \pi \times (D - d) / L$

Example

D = 300 mm
d = 150 mm
H = 250 mm

L = 261.01 mm
Ro = 522.02 mm
Ri = 261.01 mm
$\theta = 103.44°$